

Auto Grader Report

UAID: 300349320

Course: CSCE 2004

Task: lab6b

Syntax - Correct

Program source:

```
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    double num, root, error;

    cout << "Enter number:\n";
    cin >> num;

    // Square root using sqrt
    root = sqrt(num);
    cout << "sqrt(" << num << ")= " << root << endl;
    error = num - root * root;
    cout << "error= " << error << endl << endl;

    // Square root using pow
    root = pow(num, 0.5);
    cout << "pow(" << num << ", 1/2)= " << root << endl;
    error = num - root * root;
    cout << "error= " << error << endl << endl;

    // Cube root using cbrt
    root = cbrt(num);
    cout << "cbrt(" << num << ")= " << root << endl;
    error = num - root * root * root;
    cout << "error= " << error << endl << endl;

    // Cube root using pow
    root = pow(num, 1.0 / 3.0);
```

Test #1 - Correct

Test input:

123

Your program output:

```
Enter number:
sqrt(123)= 11.0905
error= -1.42109e-14

pow(123, 1/2)= 11.0905
```

Expected program output:

```
Enter number:  
sqrt(123)= 11.0905  
error= -1.42109e-14  
  
pow(123,1/2)= 11.0905
```

Test #2 - Correct

Test input:

```
-42
```

Your program output:

```
Enter number:  
sqrt(-42)= -nan  
error= -nan  
  
pow(-42, 1/2)= -nan
```

Expected program output:

```
Enter number:  
sqrt(-42)= -nan  
error= -nan  
  
pow(-42,1/2)= -nan
```

Score = 100%

Grade Summary

UAID	300349320
lab1a	100%
lab2a	100%
lab2b	100%
lab2c	100%
lab3a	100%
lab3b	100%
lab3c	100%
lab4a	100%
lab4b	100%
lab4c	100%
lab5a	100%
lab5b	100%

lab5c	100%
lab6a	100%
lab6b	100%